

☒ Easy (10 Questions)

1.

Which device operates at the network layer of the OSI model?

- A) Switch
- B) Router
- C) Hub
- D) Bridge

Answer: B

Explanation: Routers operate at Layer 3, handling IP addresses.

2.

What is the function of ARP?

- A) Translates domain names
- B) Routes packets
- C) Resolves IP to MAC
- D) Encrypts data

Answer: C

Explanation: ARP maps IP addresses to MAC addresses.

3.

Which device connects two different networks together?

- A) Switch
- B) Repeater
- C) Router
- D) Hub

Answer: C

Explanation: Routers connect different networks and route packets.

4.

A modem is used to:

- A) Convert analog to digital
- B) Manage network traffic
- C) Filter IP addresses
- D) Assign MAC addresses

Answer: A

Explanation: Modems modulate and demodulate signals.

5.

Which protocol provides reliable communication?

- A) UDP
- B) TCP
- C) ICMP
- D) ARP

Answer: B

Explanation: TCP ensures reliability and ordering.

6.

What does VPN stand for?

- A) Virtual Public Network
- B) Very Private Network
- C) Virtual Private Network
- D) Verified Packet Network

Answer: C

Explanation: VPN creates secure tunnels over public networks.

7.

Which address is used for routing on the internet?

- A) MAC address

- B) IP address
- C) Port number
- D) Subnet mask

Answer: B

Explanation: IP addresses are used for routing.

8.

Which protocol is used for error reporting?

- A) TCP
- B) UDP
- C) ICMP
- D) SMTP

Answer: C

Explanation: ICMP handles diagnostics like ping.

9.

A device that filters traffic based on rules is a:

- A) Switch
- B) Firewall
- C) Router
- D) Hub

Answer: B

Explanation: Firewalls use rules to allow/deny traffic.

10.

Which network type is used in a home?

- A) WAN
- B) PAN
- C) LAN
- D) MAN

Answer: C

Explanation: LANs are used for small geographical areas.

☒ Medium MCQs (30 Questions)

11.

Which layer of the OSI model does a router use to forward packets?

- A) Data Link Layer
- B) Transport Layer
- C) Network Layer
- D) Session Layer

Answer: C

Explanation: Routers use the Network Layer (Layer 3) for packet forwarding based on IP.

12.

Which of the following is NOT a benefit of subnetting?

- A) Reduces network congestion
- B) Increases broadcast traffic
- C) Improves network security
- D) Efficient IP address management

Answer: B

Explanation: Subnetting reduces, not increases, broadcast traffic.

13.

What is the main function of a gateway in networking?

- A) Extend Wi-Fi range
- B) Switch data within a LAN
- C) Translate between different network protocols

D) Filter packets based on port number

Answer: C

Explanation: Gateways enable communication between different protocols or systems.

14.

Which device increases the collision domains in a network?

A) Switch

B) Router

C) Bridge

D) Hub

Answer: D

Explanation: Hubs broadcast to all ports, increasing collisions.

15.

Which protocol is connectionless and suitable for real-time applications?

A) TCP

B) UDP

C) IP

D) ICMP

Answer: B

Explanation: UDP is used in real-time due to low overhead and no connection requirement.

16.

Which addressing scheme is used by Layer 2 devices?

A) IP Address

B) Port Number

C) MAC Address

D) URL

Answer: C

Explanation: Layer 2 uses MAC addresses to forward frames.

17.

Which protocol helps routers exchange information to determine paths?

- A) FTP
- B) SMTP
- C) OSPF
- D) ARP

Answer: C

Explanation: OSPF is a dynamic routing protocol that enables routers to share routing information.

18.

A default gateway is used when:

- A) Data is staying within the same LAN
- B) A packet is sent to a different IP on the same subnet
- C) The destination is on a different network
- D) ARP fails

Answer: C

Explanation: Default gateway routes packets to external networks.

19.

Which protocol dynamically assigns IP addresses to devices?

- A) DNS
- B) DHCP
- C) ARP
- D) NAT

Answer: B

Explanation: DHCP automates IP address configuration.

20.

In which layer is NAT performed?

- A) Application
- B) Transport
- C) Network
- D) Physical

Answer: C

Explanation: NAT operates at the Network Layer (Layer 3).

21.

Which type of IP address is used on the internet?

- A) Public
- B) Private
- C) Static
- D) MAC

Answer: A

Explanation: Public IP addresses are routable on the internet.

22.

What is the function of ICMP's Echo Request and Echo Reply?

- A) Domain resolution
- B) VPN negotiation
- C) Connectivity testing
- D) Encryption

Answer: C

Explanation: ICMP Echo is used by tools like ping to test network reachability.

23.

Which of the following can reduce the size of a broadcast domain?

- A) Switch
- B) Bridge
- C) Router

D) Repeater

Answer: C

Explanation: Routers break up broadcast domains.

24.

Which field in an IP packet ensures correct delivery sequence?

A) TTL

B) Sequence Number

C) Header Checksum

D) Flags

Answer: B

Explanation: TCP uses sequence numbers to maintain correct order.

25.

Which class of IP is typically used for large networks?

A) Class A

B) Class B

C) Class C

D) Class D

Answer: A

Explanation: Class A provides a large number of host addresses.

26.

The IPv6 address is how many bits long?

A) 32

B) 64

C) 128

D) 256

Answer: C

Explanation: IPv6 addresses are 128 bits.

27.

Which protocol resolves MAC addresses from IP addresses?

- A) DNS
- B) ARP
- C) ICMP
- D) SNMP

Answer: B

Explanation: ARP is used to find the MAC for a given IP.

28.

Which of the following uses port 443?

- A) HTTP
- B) FTP
- C) SSH
- D) HTTPS

Answer: D

Explanation: HTTPS uses port 443.

29.

A Layer 2 switch makes decisions based on:

- A) IP addresses
- B) MAC addresses
- C) Port numbers
- D) Protocol type

Answer: B

Explanation: Switches forward frames using MAC addresses.

30.

Which protocol is responsible for assigning IPs dynamically?

- A) ICMP
- B) FTP
- C) DHCP
- D) DNS

Answer: C

Explanation: DHCP assigns IP addresses to clients.

31.

What is the broadcast address of the subnet 192.168.1.0/24?

- A) 192.168.1.0
- B) 192.168.1.1
- C) 192.168.1.255
- D) 192.168.1.254

Answer: C

Explanation: /24 means subnet mask is 255.255.255.0, so 192.168.1.255 is the broadcast.

32.

Which device can translate public IPs to private IPs?

- A) DNS
- B) Switch
- C) NAT router
- D) Firewall

Answer: C

Explanation: NAT performs address translation.

33.

Which of the following represents a loopback IP address?

- A) 127.0.0.1
- B) 192.168.0.1
- C) 10.0.0.1

D) 255.255.255.0

Answer: A

Explanation: 127.0.0.1 is used for loopback testing.

34.

Which of the following best describes tunneling in VPN?

- A) Routing packets through satellite
- B) Encrypting DNS requests
- C) Encapsulating data within another protocol
- D) Blocking unsolicited packets

Answer: C

Explanation: Tunneling encapsulates traffic inside another protocol.

35.

Which network device uses NAT functionality by default in home setups?

- A) Switch
- B) Modem
- C) Home Router
- D) Bridge

Answer: C

Explanation: Home routers usually perform NAT to allow multiple devices to share a single public IP.

36.

What kind of IP address is 192.168.0.10?

- A) Public
- B) Private
- C) Multicast
- D) Broadcast

Answer: B

Explanation: 192.168.x.x is a reserved private IP range.

☒ Hard MCQs (10 Questions)

41.

Which routing protocol supports VLSM and is classless?

- A) RIPv1
- B) RIPv2
- C) BGP
- D) IGRP

Answer: B

Explanation: RIPv2 supports classless inter-domain routing and VLSM.

42.

What does TTL in an IP packet prevent?

- A) Packet duplication
- B) Packet spoofing
- C) Routing loops
- D) Fragmentation

Answer: C

Explanation: TTL ensures that packets do not circulate indefinitely.

43.

In CIDR notation, what does /20 represent?

- A) Subnet mask of 255.255.255.0
- B) Subnet mask of 255.255.240.0
- C) 16 bits of network

D) 20 hosts per subnet

Answer: B

Explanation: /20 = 255.255.240.0

44.

Which IP class offers the smallest number of usable host addresses?

A) Class A

B) Class B

C) Class C

D) Class D

Answer: C

Explanation: Class C provides 254 usable addresses per subnet.

45.

What is the function of the routing table in routers?

A) Manage DHCP leases

B) Encrypt data

C) Determine the best path for a packet

D) Assign MAC addresses

Answer: C

Explanation: Routing tables store paths to various network destinations.

46.

Which IPv6 address type is equivalent to a broadcast in IPv4?

A) Anycast

B) Multicast

C) Unicast

D) Link-local

Answer: B

Explanation: IPv6 uses multicast instead of broadcast.

47.

What happens if a router does not have a matching route for a packet?

- A) It broadcasts the packet
- B) It returns it to the sender
- C) It drops the packet or uses a default route
- D) It forwards to the DNS server

Answer: C

Explanation: If no route is found, the packet is either dropped or sent via the default route.

48.

What distinguishes IGP from EGP?

- A) IP version used
- B) Layer of operation
- C) Whether routing is internal or external
- D) Port number used

Answer: C

Explanation: IGP is for internal routing; EGP (like BGP) is used between different autonomous systems.

49.

Which IP address range is reserved for APIPA?

- A) 10.0.0.0 – 10.255.255.255
- B) 172.16.0.0 – 172.31.255.255
- C) 169.254.0.0 – 169.254.255.255
- D) 192.168.0.0 – 192.168.255.255

Answer: C

Explanation: APIPA assigns addresses from the 169.254.x.x range when DHCP fails.

50.

Which of the following is NOT a valid private IP address?

- A) 10.0.0.5
- B) 192.168.100.1
- C) 172.20.5.5
- D) 8.8.8.8

Answer: D

Explanation: 8.8.8.8 is a public Google DNS IP.